

DAY 6

Lookup Functions

Lookup functions ek table me se value dhundhkar doosri jagah laate hain. Do tables ko jodne ke liye yeh sabse zyada use hone wale functions hain.

VLOOKUP aur HLOOKUP

- VLOOKUP: vertical (column wise) table me left column me dhundhta hai aur right side se value lata hai.
- HLOOKUP: horizontal (row wise) table ke liye.
- Last argument: FALSE ya 0 = exact match, TRUE ya 1 = approximate match (sorted data me).
- VLOOKUP sirf left se right dhundh sakta hai. Lookup value hamesha pehle column me honi chahiye.

XLOOKUP (Modern Function)

- VLOOKUP se behtar: kisi bhi direction me dhundh sakta hai, default exact match hai.
- Not found ke liye seedha message de sakte ho, IFERROR ki zaroorat nahi.

INDEX aur MATCH

- MATCH: value ki position (number) batata hai. INDEX: position par ki value deta hai.
- INDEX + MATCH milkar VLOOKUP ka flexible option banate hain (left lookup bhi possible).

Common Errors

- #N/A: value nahi mili. Extra spaces ya text/number mismatch check karo.
- #REF!: col_index_num table ke columns se bada hai.

Important Formulas / Shortcuts

Formula / Shortcut	Kaam (Purpose)
<code>=VLOOKUP(A2, \$F\$2:\$H\$20, 3, FALSE)</code>	3rd column se exact match
<code>=HLOOKUP(A2, \$B\$1:\$H\$3, 2, FALSE)</code>	Row 2 se value
<code>=XLOOKUP(A2, F2:F20, H2:H20)</code>	F me dhundo, H se lao
<code>=XLOOKUP(A2, F2:F20, H2:H20, "Not Found")</code>	Na mile to message
<code>=MATCH("Rahul", A2:A20, 0)</code>	Rahul ki position
<code>=INDEX(C2:C20, 5)</code>	C column ki 5th value
<code>=INDEX(C2:C20, MATCH(A2, B2:B20, 0))</code>	INDEX-MATCH combo
<code>=IFERROR(VLOOKUP(...), "Not Found")</code>	Error handling

Practice Tasks

1. Ek Product table (ID, Name, Price) banao aur doosri sheet me ID daalne par Name aur Price laao.
2. Wahi kaam XLOOKUP se karo.
3. INDEX-MATCH se ek left-side lookup karo.
4. Marks ke basis par grade approximate match VLOOKUP se nikalo.

Pro Tip: Lookup table ko absolute reference (\$) me lock karo, warna formula neeche copy karne par range khisak jayegi.